# Startup Ecosystem Around Digital Life Sciences

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## 1 The Digital Revolution and Startups in Life Sciences

Medical needs drive the global life science sector for still life-threatening diseases like cardiovascular diseases, cancer, and global pandemics. On one side, digital solutions for medical needs are available. On the other side, there are still considerable differences in the maturity of implementing the latest technology-based digital solutions on a global scale.

According to Schwab (2017, p. vii), the three previous industrial revolutions have created major societal change and opportunities. Nevertheless, the Fourth Industrial Revolution is a transformational phase, driven by rapidly changing technologies.

From the historical perspective, every industrial revolution brought abrupt and radical change. The first revolution brought the invention of steam and water power.

The second revolution was characterized by the advent of electricity, mass production, and the assembly line. The Third Revolution is called the computer or digital revolution because of the invention of semiconductors, personal computing, and the internet (Schwab, 2017).

One of the most promising biotechnology former startups, THERANOS, which pretended to revolutionize personalized medicine by delivering hundreds of analyses from just a few drops of blood, had raised many questions about venture funding and unfulfilled technology application promises. THERANOS' diagnostic technology was based on absorption spectrometry, microfluid systems, computer-assessed methods, and medical information systems. Despite the fraud and well-documented scandal, this former digital healthcare startup was at the time of inception and over the first year's part of the digital revolution in the Life Sciences.

Nevertheless, this research focuses on successful, purpose-driven startups based in different parts of the world, but with a clear focus on Germany. In this research,

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the evolution of digital health startups, their role in the healthcare ecosystem, and different evolving categories are under investigation. The focus is on life science startups from the last decade. Hence, some so-called "first movers," which do not have the startup status anymore, are mentioned to provide a holistic picture from the European and US perspective. At this point, we want to mention that our study has a blind spot on Asia. This is due to the fact that Asia plays a minor role in the field of life science startups. For more details, please read Sect. 3.

This chapter sheds light on the application of digital technologies in the life science sector. The Digital Revolution, also known as the Third Industrial Revolution, is based on the invention of the internet. Today, the Fourth Industrial Revolution is in full bloom. Based on Schwab (2017), this revolution is characterized by a transformation of humankind. Marc R. Benioff illustrated a picture of our future in his foreword to Klaus Schwab's book:

> In the coming decades, the technologies driving the fourth industrial revolution will fundamentally transform the entire structure of the world economy, our communities and our human identities. (Schwab, 2017, p. viii)

Startup organizations are the driver in many sectors; in particular, the life science industry is benefitting from the impact of startup and SME organizations. The best example from the life science industry is the growing importance of biotechnology companies, which have become the "research laboratories" for big pharmaceutical organizations.

First, no commonly agreed definition of the term "startup" exists. Nevertheless, this research is investigating in young organizations fulfilling the following criteria based on (Bormans et al., 2021):

- Younger than 10 years
- Innovative product and/or service, and/or business model
- Intention to grow (number of employees, turnover, markets)

Second, we have added the criteria that the startup has implemented digital technologies to their business model.

The original term 'ecosystem' or 'ecological system' describes all organisms, their physical environment, and how they interact (Chapin, 2011). In contrast to this definition, this research study investigates digital startups embedded in social, commercial, and technological driven ecosystems. One important phenomenon of these ecosystems are multisided platforms, which are created or, to some extent, evolving around the innovative product, services, or technologies the startups bring to the market. Many diverse digital health stakeholders need to be addressed. Therefore, digital healthcare ecosystems are complex and characterized by an aversion to risk, highly regulated markets (EMA, FDA), cost, and data-driven strategies (Teece et al., 2022).

Hence, from our data collection of 30 companies, only a recent perspective can be provided. The following five clusters have emerged from this investigation:

- Early detection and diagnosis
- Treatment of diseases
- Patient empowerment
- Services for physicians and insurance provider services
- Technology applications

Our sample of companies is indeed limited. Our research project was limited in the perspective of time and access to data. It lies in the nature of startup companies that not all data are available, especially during the founding phase. Our sample is also limited due to the numerous platforms, databases, and divers sources of data. Therefore, we started with a long list of startups, did the clustering, and in the next step, included companies, with internet presence and product information. This led us to the recent perspective of the five clusters and the presented startups.

## 2 Digital Health Stakeholders

The life science sector is characterized by the orchestration of many different stakeholders, which bring in different interests and value propositions. Nevertheless, this complex ecosystem is evolving around patients and citizens in general. All parties, like companies, policy makers, education and research, funders, the third sector and health and social care providers, create value for our society's patients and individuals. Taking those stakeholders' social, ecological, ethical, and economic perspectives into account leads to a complex picture of the ecosystem for startups in the life sciences. From the economic perspective, startups' role and influence as stakeholders should not be underestimated.

This graphic illustrates the complexity and diversity of all involved stakeholders, but how they interact, collaborate, and compete needs a deeper understanding. Hence, this graphic demonstrated that startups are important industry stakeholders. The startups, SMEs, and large corporates are defined by pharma companies, biotech companies, medical device companies, diagnostic companies and IT companies focusing on medical services. This complex perspective is reflected in our research about startups and their ecosystems.

## 3 Digital Startups: The Last Decade

According to a study by the data platform Statista, the financing of US digital health startups increased by more than 2600% between 2011 and 2021 (Statista, 2022a). While in 2011 a total of 1.1 billion US dollars were spent to promote startups in this area, in 2021 it was already 29.2 billion US dollars. Especially in the years between 2017 and 2021, market growth can be observed. Figure 1 illustrates this development.

In addition, the study shows that in 2020 the USA had the largest share of promising digital health startups at 76%. By comparison, second-ranked China

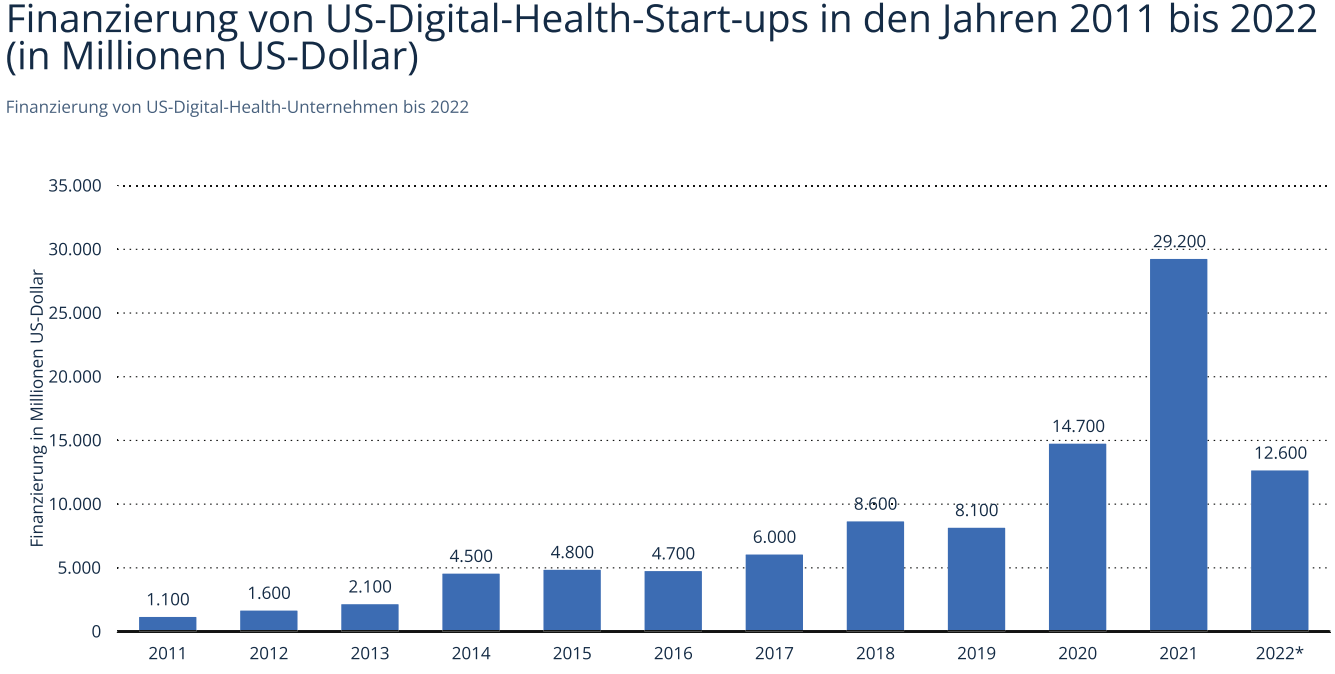



**Fig. 1** Financing of US digital health companies until 2022 (Source: Statista, 2022a)

only accounts for 4% of all startups in this area (Statista, 2022a). This insight implies that the USA is an unrivaled leader in the foundation of digital health startups. The 150 most promising digital health startups by country are illustrated in Fig. 2.

A similar conclusion can be reached by looking at lists of the best-funded startups of different years. Eight of the ten most funded companies in 2019 are located in the USA (Statista, 2022b). As shown in Fig. 3, only ranks 3 and 5 are occupied by non-US companies: WE DOCTOR (China) and BABYLON HEALTH (UK).

We take this as an opportunity to first divide these ten companies into clusters, whereby each of these clusters is assigned to a different type of product and/or service. According to a forecast by the consulting firm Roland Berger, the turnover of digital health companies on the global market will rise to one trillion euros by 2026. Of these, 5.9% will be generated in Germany.

Summarizing the findings of Figs. 1 and 3, we conclude that Germany will play a rather subordinate role in the global digital health market. For this reason, data suggested that the initial clustering for different services and/or products does not sufficiently cover the German market. We will therefore expand the first clustering by including the most important startups from Germany from 2015 and the German digital health startups founded in 2020/21. Finally, we compare our cluster formation with a distribution of the segments of all the most promising digital health startups from 2020.

## 4 Emerging Clusters in Digital Life Science Startups

The intensive research study led to five clusters of digital health startups based on the specific services, products, technologies, and business models. We approached the data open-mindedly, to not restrict the outcome and the new perspective these clusters provide. We searched for patterns and comparability, to provide a logical structure for the evolving cluster. After final consideration, the clusters were ordered according to early detection and diagnosis; treatment of diseases; patient empowerment; services of physicians and insurance providers; and technology applications. The last cluster, technology application, is an important stand-alone cluster, since we have identified startups, which have implemented the new digital technologies or the new combination of digital technologies to their products and services. Nevertheless, our evolving clusters are not claiming to be all-encompassing or universal. This provided a recent picture of newly founded startups and startups that have changed and revolutionized the application of digital technologies in the life sciences.

### 4.1 Early Detection and Diagnosis

The diagnostic industry is one of the prominent examples of the early adoption of telemedicine and telehealth, in other words, digital health applications. Taking the medical discipline pathology as an example, telemedicine is a cornerstone for tissue diagnosis and has been used for decades by pathologists to discuss critical patient

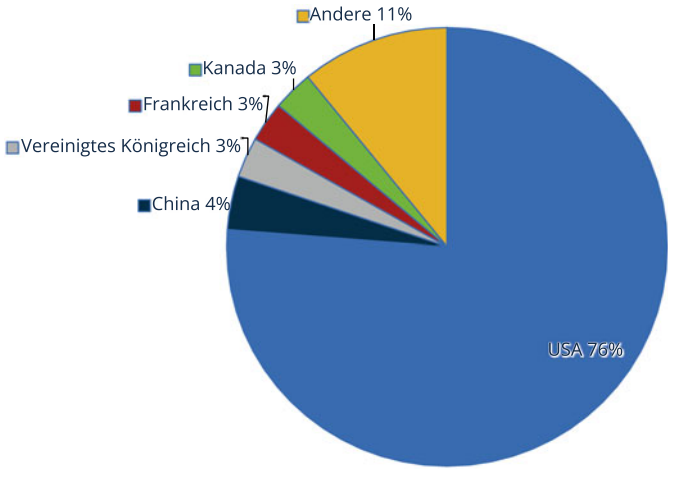



**Fig. 2** Distribution of the 150 most promising digital health startups by countries in 2020 (Source: Statista, 2022b)

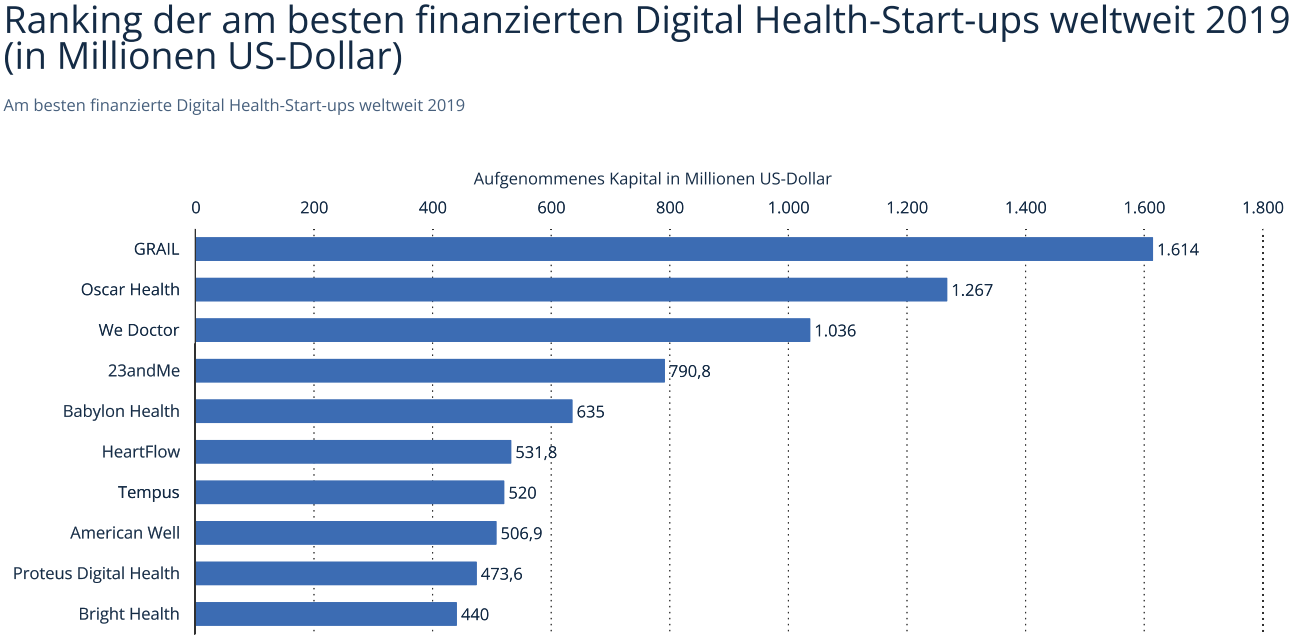



**Fig. 3** Ranking of the best financed digital health startups worldwide in 2019 (in Million USD) (Source: Statista, 2022c)

cases. Here, data from immunochemistry, histology, and microscopy are exchanged, analyzed, and evaluated over long geographical distances.

Nevertheless, today advanced digital technologies provide much more opportunities for early detection and precise diagnosis of many diseases. The following startups are representative examples for this cluster, with the limitation that numerous other active startups are in the market or recently founded.

23andMe, founded in 2006 by Anne Wojcicki in San Francisco, USA, is one of the first movers in providing DNA analysis to private people. Their business model is based on collecting data from millions of people with different ethnical backgrounds. The data analysis provides information about a person's heritage and their predisposition to specific diseases. Precisely this part of the analysis, which was critical from the ethics perspective, led to the fact that the FDA had abandoned 23andMe from the market in the first instance but received the first-ever authorization for direct-to-the-consumer cancer risk analysis test kits in 2018.

As of today, the company is successful (12 million test kits sold) and one of the role models in early disease detection and diagnosis. To some extent, 23andMe is also relevant for the cluster patient empowerment since for the first time, people could gain insights into their DNA analysis without consulting a doctor (23andMe, 2023).

HEARTFLOW was co-founded by Christopher Zarins and Charles Taylor in 2007 in Mountain View, in the USA. This early adopter of artificial intelligence (AI) in precision heart diseases is today very well established in the US Market. Their AI algorithm-based product HEART FLOW is implemented in 80% of the top 50 US heart hospitals. Especially in the field of cardiovascular diseases, early diagnosis can save lives. With their personalized 3D model of every single patient's heart, the diagnosis and potential treatments are much more effective and precise (Heartflow, 2023).

GRAIL, co-founded by Alex Aravanis, Jeff Huber, and Mostafa Ronaghi in 2015 in San Francisco, USA, has developed a diagnostic test kit to detect early cancer. Their product GALERRI is a multicancer detection kit to provide analysis from just one bold sample. Their approach has been further developed to screen for up to 50 types of cancer, for which there are usually no early screenings. They obtain their results by analyzing cell-free DNA samples for specific methylation patterns. Specific algorithms analysis supports their results. With their approach, GRAIL provides important early-stage analysis on one side and significant patient data collections on the other. The company was acquired by ILLUMINA (Grail, 2023).

One of the US's largest patient data platform organizations is TEMPUS, founded by Eric Lefkofsky in 2015. The company provided multisided services to patients, diagnostic laboratories, and insurances. Their big promise is to become the world's largest clinical and molecular data library. Based on their special data collections, they can build virtual replicas of a patient's DNA or RNA. They have adapted AI and machine learning of providing their customers with the most precise data about their health status (Tempus, 2023).

SIGN HEALTH TECHNOLOGIES was co-founded in Germany by Dennis Dickmann, Kai Kölsch, and Thassilo Alexander Vogt in 2020. The company has

developed a medicinal product, a bandage with temperature sensors for fever measurement. The temperature is measured every 10 min. An algorithm analyzes the results, which are visualized in a mobile application. This technology should support controlling the health status of nurses, social workers, and other personnel, where early identifying of changes in their health status is important (Sign Health Technologies, 2023).

Another representative example for a startup, focusing on the early detection of biomarkers related to heart diseases, is X-CARDIC. The company was co-founded in 2021 by Oliver Höppner and Dr. Alexander Meyer in Berlin, Germany. The startup has applied AI to its first product X-C-BLEEDING to identify the potential for postoperative bleeding. This condition is still a cause for numerous complications related to the heart surgeries. Up to 100,000 patients/year suffer from severe postoperative complications. Therefore, the company is addressing an unmet medical need. Their research studies have provided convincing data for the use of machine learning applications in the field of cardiovascular medicine. This digital health application addresses the medical need from the patient perspective, and the cost pressure of hospitals in general, particularly growing intensive care concerns (X-Cardic, 2023).

According to their internet presence, the startup CURE CURVE has applied breakthrough technologies to "Detect, Prevent, and Solve the toughest healthcare challenges." Their AI-based App ELINA provides health information and personalized advice for patients with cardiovascular diseases. The startup has developed an all-in-one solution for the life of the patients, from tips for personalized exercises, drug treatment schedules, and healthy recipes to reminders for important appointments (Curecurve, 2021).

## 4.2 Treatment of Diseases

The largest cluster we have found includes startups that deal with the treatment of diseases and offer digital solutions. As part of this analysis 11 startups which formed the treatment of diseases cluster will be presented here. Due to the size of this cluster, it was decided to first introduce the three most important startups in the world and then discuss German foundations.

The company WE DOCTOR is a spin-off of the Chinese healthcare group TENCENT. The company has brought together 7000 health experts from China and offers a worldwide 24/7 support. Services include psychological support, online health enquiry service, prevention guidelines, and real-time pandemic reports. The offers are provided via service hotlines and chat services (e.g., "WeChat"). The parent company TENCENT was founded in 1998 in Shenzhen, China, and has been listed on the Hong Kong Stock Exchange since 2004 (Tencent, 2023). The spin-off WE DOCTOR was launched in 2010 and is headquartered in Hangzhou, China (CB Insights, 2022).

In 2012, the startup BABYLON HEALTH was founded in London by the British-Iranian investment banker Ali Parsa. It offers remote contacts between

doctors/healthcare professionals and patients. These contacts are offered via test and video messages. BABYLON HEALTH has employed over 2100 people in 2021, generating more than 300 million US dollars in revenues. Key partners include the National Health Service (NHS) and the Consumer Health Service Higi (Babylon Health, 2023).

Co-founded in 2012 by Nitin Nanda and Vikram Marla, ALIGNED TELEHEALTH offers SaaS-based telemedicine services. It was sold to American Well Corporation (called Amwell since 2020) in 2019. This makes the group, founded in 2006 by brothers Ido and Roy Schoenberg, one of the most important providers in the field of telemedicine. The company employs approximately 800 people and is funded with more than $500 million. Key investors include Allianz Insurance, Philips, and Teva Pharmaceuticals (Amwell, 2023).

All three companies have in common that they provide online platforms where patients can digitally connect with doctors and medical staff. Such platforms thus represent the most important digital innovation in the field of healthcare. Those platforms are available for all common digital devices like smartphones, tablets, and personal computers. Companies have also been founded in Germany to establish digital contact between physicians and patients.

The coaching company HAPPY HABITS SOLUTION, founded by Nikita Fahrenholz and headquartered in Berlin, offers the app ACTIVIO, a service with which experts from different directions can get in touch with interested parties. In this way, the experts become coaches. In 2020, Fahrenholz received EUR 8.5 million in funding from investors HV Capital and Cavalry Ventures (Heuberger, 2020). The health coaches are offering services in the field of health topics such as nutrition and psychology.

Co-founded in 2016 by Max Michels, Benjamin Pochhammer, and Max von Waldenfels, the company GOREHA, better known as CASPER HEALTH, works closely with rehabilitation clinics. It offers app-based services in the field of occupational therapy and physiotherapy. In addition, sports science and psychological counseling are part of the company's digital service offers. The Berlin-based company already has more than 40 employees (Caspar, 2023).

In 2016, Gloria Seibert and Clemens Kofler founded the company TEMEDICA and received initial financing in 2017. Further financing followed, enabling the company to grow to 70 employees by 2020. The company's goal is to improve patient care by collecting patient data and making it available for research. In return, the patient is given the opportunity to get in touch with experts in the medical field with the help of app applications. For example, the company provides special app offers for patients with multiple sclerosis or psoriasis. Also offers for patients suffering from Crohn's disease are in development (Temedica, 2023).

The startup SELFAPY, co-founded in 2016 by Katrin Bermbach, Nora Blum, and Farina Schurzfeld, offers online courses for treating depression, eating disorders, and other psychological disruptions. Since introducing the Digital Supply Act 2019, these digital services have been available free of charge on prescription and are accepted by all statutory health insurance companies (Gesetzliche Krankenversicherungen—GKVs). SELFAPY employs over 100 people and more

than 25 psychologists monitor clients' mental health (Selfapy, 2023). This startup, founded by three young women, is an excellent example of how digital technologies can fill the gap between patients' medical needs and the real availability of these types of services. Like ADA HEALTH, SELFAPY is one of the first mover in digital health services in Germany (Selfapy, 2023).

The startup company ELONA HEALTH, co-founded by Alexander Braden, Magnus Schückes, Johanna von Lobenstein, Peter Neudeck, and Leon Hillebrand, was awarded with the founding prize of the Ministry of Economic Affairs and Climate Protection in 2021. It offers an app with which patients undergoing psychotherapy can independently plan their upcoming therapy steps. Other apps, with which patients can be prepared for therapy or relapses after therapy can be avoided, are already in preparation. The company already has 20 employees (Elona, 2023).

As these examples show, online platforms and apps that connect patients with medical staff, accompany patients in therapies, or inform and advise on diseases are a booming market, which is reflected in the growing number of startup organizations. These digital technologies are used worldwide and already provide jobs for thousands of people. It can be assumed that this market will continue to exist and grow considerably in size. The application of digital technologies is an important driver for providing value to patients' medical needs and the whole society from the perspective of preventing diseases.

## 4.3 Patient Empowerment

Patient empowerment is a vital cluster with growing impact on the digital health market. Patient empowerment does not only stand for the patient as the driver for digital applications but also for the movement of the informed patient, which started two decades ago. An example for this is the platform PATIENTSLIKEME, co-founded by Jamie Heywood, Jeff Cole, and Ben Heywoodin 2004 in the USA. This was one of the first patient's communities, founded by people with rare diseases, with the purpose to get access to the best treatments available, based on international research data. Today, the community has more than 850,000 members worldwide (Patientslikeme, 2023).

ADA HEALTH was co-founded by Claire Novorol, Martin Hirsch, and Daniel Nathrath in 2011. From the perspective of the highly regulated German healthcare market and the slow introduction of new technologies to this critical sector, ADA HEALTH was one of the first movers in adapting artificial intelligence to their service solutions. At this time, providing medical diagnosis not in person but via the internet was a milestone in digital health. Against all odds, the company managed to convince the authorities that their virtual diagnosis, based on the patients' symptoms, can become a medicinal product and therefore introduced directly to the customer. ADA HEALTH is today one of the leading service providers of end-user self-assessment web applications. Individuals can address their symptoms via an App and get profound diagnosis and advice for the next

steps to become healthy again. In cases of serious diseases, the patients get immediate access to physicians.

Based on the AI and constant clinical research, the company is providing their services not only B2C but also B2B. With more than 13 million users worldwide, and nearly 30 million assessments, ADA HEALTH has provided evidence for the power of the individual. All these users have decided to take care of their health out of their own motivation. With 50 healthcare experts, physicians, QM experts, medical engineers and healthcare researchers, and offices in Berlin, London, New York, and Toronto, the company is on the path to becoming a global player.

Therefore, ADA HEALTH is a German role model for digital startups, which have enabled the empowerment of patients (Ada Health, 2023).

The following examples are startup companies which provide holistic services, information, and products to individuals in specific life situations, like pregnancy and motherhood, patients with specific conditions, and couple partnerships.

KINDERHELDIN was founded in 2017 by Dr. Paul Hadrossek and Fabian Müller, in collaboration with the digital health platform Heartbeat Labs. The platform provides health information and services to pregnant women from the beginning of their pregnancy to the early years of their child. Services like video chats with a midwife are very supportive services, especially for young parents. Since 2019, KINDERHELD has had contract collaborations with 50 GKVs in Germany. This type of services allows women and their families to empower themselves in the period of motherhood and parenting (Kinderheldin, 2023).

The digital platform startup CARA CARE was founded by Jeseja Brinkmann and Andre Sommer in 2016 in Germany. Their web applications focus on advice for the holistic treatment for chronic gastrointestinal issues. With a personalized digital diary, which is adapted for every single person, CARA CARE is a tool for these patients to manage their daily life with the illness. Three approved medicinal products for the following conditions are available: irritable colon disease syndrome, chronic irritable colon disease, and irritable stomach disease. All information, suggestions, and recommendations are based on clinical research and expert knowledge. CARA CARE has developed an adaptable diary for every patient's specific needs. Their e-health solution has significantly enhanced how the patients can deal with their condition. CARA CARE is a collaboration with physicians and the GKVs. The app can be prescribed via a recipe. Patients are provided with the right tools and information to influence their conditions in a self-empowered and personalized manner (Cara Care, 2023).

HEALTH CATERS was co-founded in Germany by Tatyana Eliseeva and Lilia Kruse in 2021. The startup is still early stage and provides a holistic package of different products to enable organizations to allow their employees to measure their health data. The offered digital health management is addressing the need for informed decisions regarding a healthy lifestyle. Employees can use these digital services to improve their health, vitality, and longevity. The B2B business model addresses the organizations' need for preventive health management inside of their organizations (Health Caters, 2023).

Employees are empowered to make healthy decisions in their working environment. HEALTH CATERS are at the forefront of the movement that health care is no longer private. It is an essential factor in the daily world of business.

UMAVERSUM is another digital service platform providing information and advice in the life-changing time of pregnancy. The startup was co-founded in Germany by Mirjam Peters, Elena Kirchner, and Jonah Polack in 2021. Their B2C business model focuses on providing scientific and personalized data and information to pregnant women over the whole period of pregnancy. A study about the quality of the UMA app has shown that pregnant women felt more confident regarding their recent health condition. They experienced better sleep and less depression. The positive effect of the app was approved. This digital solution will enable the women and their families to influence their pregnancy in the different stages in a positive and self-empowered manner. The information provided is important to cope with this new period of life. Women can gain more security regarding their health decisions; they gain personal control over the situation and practical help and orientation (Umaversum, 2023).

The startup RECOUPLING was co-founded in Germany by Jaane Henning, Dr. Mandy Hecht, and Johanna Klapprott-Lubig in 2021. The web application was developed to help couples cope with conflicts and maintain a healthy, interesting, lucky partnership. This type of service is well addressed to the needs of the so-called digital natives generation, which is technology affine and interested in optimization of all areas in their lives, including partnerships.

All these startups have in common that their products and services enable patients or individuals in specific stages of life to make informed, self-empowered decisions for a better and healthier lifestyle. By providing medical information, like personal diagnosis, relevant latest research results, and recommendations, i.e., to pregnant women, changing circumstances can be managed without numerous visits to their doctors.

## 4.4 Services for Physicians and Insurance Providers Services

In this cluster, companies that provide services for doctors will be presented.

A common problem among general practitioners is the correct billing of their services with German health insurance companies. These must comply with the uniform assessment standard of the National Association of Statutory Health Insurance Physicians. To make this often complex and laborious process easier for doctors and thus increase revenues, SYSTOLICS, co-founded in 2020 by siblings Lara and Uli Kraulfuss in Würzburg, offers the SYSTOLICS OPTIMIZER. This cloud-based service regulates and optimizes billing for medical practices. In addition, the company offers controlling services such as benchmark analyses. The company is a spin-off of the Fresenius University of Applied Sciences (Systolics, 2023).

It is a challenge for doctors and hospitals to provide patients with information. To better meet this challenge, the startup SETHEALTH offers the service to integrate

medical images into iOS, android, and web applications. This allows patients to be provided with images that can be used to explain their diseases and suggest therapies. In addition, the company is working on a service that will bring doctors into contact with bioengineers. Due to this collaboration, companies should be offered the opportunity to manufacture medical products tailored to the customer faster and cheaper. The startup was founded in Berlin in 2020 by Manu Martinez-Almeida (SetHealth, 2023).

This sub-category provides insights into the development of service applications, focusing on the demands of insurance providers, and targeting unfulfilled patients' needs. Interestingly, we did not find startups addressing solutions to statutory health insurance companies (GKV)). There is only informal information available that the GKVs, particularly in Germany, have recently adopted innovation processes and idea competitions insight of their organizations.

On the mission to reinvent health insurance, the US company OSKAR HEALTH was co-founded by Joshua Kushner, Mario Schlosser, and Kevin Nazemi in 2012. The multisided platform business model is enabling OSKAR HEALTH to provide patients with better digital solutions for their immediate needs, like consultations with a primary care provider (PCP). The company has implemented new technologies to provide better care services. Their OSCAR App gives access to health plans, access to prescriptions, a review of laboratory results, and virtual access to the patient's personal ID cards. As of January 2022, the insurance company counts approx. one million members. They are active in 20 states of the USA. They provide services to healthy individuals, patients and families, healthcare providers, and brokers. The digital platform enables this insurance company to offer their members high standards of medical care, but also services around wellness and disease prevention (Oskar Health, 2023).

Startups like BRIGHT HEALTH are impressive examples from the US insurance market. The company was founded in 2016 by Bob Sheehy, Kyle Rolfing, and Tom Valdivia. Over the years, BRIGHT HEALTH became one of the major players in the application and practical fulfillment of the "Affordable Care Act (ACA)" announced in 2010 by former President Obama, also known as "Obamacare."

Due to intensive M&A activities, investments, and growth, BRIGHT HEALTH reached unicorn status in 2020 (Miller, 2020). With this recent evaluation, the company is one of the most influential US insurance and health plan providers for patients, doctors, and hospitals. Their service platform has technologies like user interfaces, electronic medical records, telemedicine, and data analytics modules incorporated. More targeted, patients' specific value-based delivery of care is the purpose of their business model (Brighthealthgroup, 2023).

## 4.5 Technology Applications

By analyzing the startup companies presented so far and their evolving clusters, our data collected provided evidence that startups that develop new technical devices are strongly underrepresented. Therefore, companies whose products represent new

technical achievements or applications will be presented in this cluster. The technology applications cluster consists of four companies specializing in developing such technologies. The focus here was on German companies that were founded recently, from 2020 to 2022. Thus, this cluster represents a highly topical excerpt from the German startup scene for new technology applications in the medical field.

The startup EXAGOL MEDTECH, founded in Hamburg in 2020, produces automatic venepuncture machines. The company uses machine learning algorithms for better pinpricking accuracy and ultrasonic scanning to detect optimal puncture sites. In 2021, the company was awarded second prize in a competition organized by ARC Sheba Medical Centres. The eight-person team, which includes medical technician Van Nguyen, software and hardware developer Lawrence Netzer, and product manager Martin Bukowski, is working to solve the problem of often difficult blood samples from patients. Full market entry is expected in 2024 (Excagol Medtech, 2023).

A modern method to visualize molecules and low-molecular-weight substances is Raman spectroscopy. It is used to analyze antibiotic concentrations in aquatic environments and determine metabolites in body fluids (Lussier et al., 2019; Wali et al., 2019). In order to improve the sensitivity and reproducibility of SERS substrates used in Raman spectroscopy, the startup NANOSTRUCT, co-founded in 2021 by Thien Ahn Le, Enno Krauss, Henriette Maaß, and Kai Leibfried, produces ultra-precise gold nanostructures. These consist of a single-crystal gold film with an extension of only 200 μm and can be produced identically in large quantities. The company emerged from the University of Würzburg and is funded by the Federal Ministry for Economic Affairs and Energy and the European Union (Nanostruct, 2023).

The company BEARCOVER produces the robot OSCAR. This night worker, equipped with a radar sensor, is intended to monitor hospital patients without entering the rooms. In this way, OSCAR is to support the team of nurses and caregivers. The robot is controlled by an app and checks the patients every 15–20 min. The startup was co-founded by Ben Duffy, Fernando Torales, and Daniel Flynn in Berlin in 2020 and is supported by KALUCARE and the Berlin School of Economics and Law. Here, we see the application of robotics to address the need for more personnel, especially in the field of care for the elderly.

In order to support people with neuronal diseases, the company CEREGATE, founded in Munich in 2019, is developing imaging techniques and new therapies. For this purpose, information from the environment of patients is to be transferred to previously implanted electrons. Later, it is measured which information has arrived in the patient's brain. In other words, CEREGATE fuses clinical neuromodulation with brain–computer interfacing. The now 14-person team around founder Balint Varkuti, who previously worked in the neurotechnology industry, consists of experts from both disciplines. The firm receives support from 468 Capital, re.Mind Capital, and other investors (Ceregate, 2023).

## 5 Conclusions

Studying the startup ecosystem around digital life sciences has provided the authors with recent insights into the latest technology developments, applications, and maturity of digital technologies. The sample of 30 organizations has illustrated how the Fourth Digital Revolution has influenced the healthcare sector with all stakeholders. We have found all digital health stakeholders to some extent involved in the different clusters of startups. Even if this research has the limitation, that not all geographical regions could be covered, and many life science startups are still "under the radar," we have been able to provide a recent perspective on how digital solutions are reaching and empowering patients and citizens.

Hence, breakthrough technologies like machine–human interfaces, implantable robotics, health services delivered via the metaverse, and digital twins are still underrepresented in technology applications. But we expect to see much more of these technologies in the near future.

## Appendix: Company Overview with Founding Year, Products, and Contact Information

| Cluster | Founding | Company | Profile/Product | Company link |
|---|---|---|---|---|
| Early detection and diagnosis | 2006 | 23andMe | DNA analysis | https://www.23andme.com/ |
| Early detection and diagnosis | 2007 | HeartFlow | AI, precision heart diseases | https://www.heartflow.com/ |
| Early detection and diagnosis | 2015 | Grail | Multicancer detection kit | https://grail.com/ |
| Early detection and diagnosis | 2015 | Tempus | AI diagnostics | https://www.tempus.com |
| Early detection and diagnosis | 2020 | Sign Health Technologies | Bandage with temperature sensors | https://www.signsdegree.com/ |
| Early detection and diagnosis | 2021 | X-Cardiac | AI, precision heart diseases | https://www.xcardiac.com |
| Early detection and diagnosis | 2021 | CureCurve | AI health information | https://curecurve.de |
| Treatment of diseases | 2004 | WeDoctor | Hotline, chat services | https://www.tencent.com |
| Treatment of diseases | 2012 | Babylon Health | Tests, video messages | https://www.babylonhealth.com |
| Treatment of diseases | 2012 | TeleHealth | SaaS-based telemedicine | https://www.business.amwell.com |
| Treatment of diseases | 2020 | Happy Habits Solution | Coaching app | https://www.actio.com/ |
| Treatment of diseases | 2016 | Goreha | App-based physiotherapy | https://www.caspar-health.com |

(continued)

| Cluster | Founding | Company | Profile/Product | Company link |
|---|---|---|---|---|
| Treatment of diseases | 2016 | Temedica | Data collecting app | https://www.temedica.com |
| Treatment of diseases | 2016 | Selfapy | Online courses, treatment depression | https://www.selfapy.com |
| Treatment of diseases | 2021 | Elona Health | App for psychotherapy assistance | https://www.elona.health.com |
| Patient empowerment | 2004 | PatientsLikeMe | Patience community platform | https://www.patientslikeme.com/ |
| Patient empowerment | 2011 | Ada Health | Self-assessment platform | https://ada.com/de/ |
| Patient empowerment | 2017 | Kinderheldin | Pregnancy support app | https:/kinderheldin.de |
| Patient empowerment | 2016 | Cara Care | Personalized treatment app | https://pages.cara.care/ |
| Patient empowerment | 2021 | Health Caters | Healthmanagement B2B | https://www.healthcaters.ai/ |
| Patient empowerment | 2021 | Umaversum | Pregnancy supprt app | https://www.uma-app.com/ |
| Patient empowerment | 2021 | Recoupling | App for couples | https://www.recoupling.de/ |
| Services for physicians and insurance providers | 2020 | Systolics | Cloud-based IT solution for medical bills | https://systolics.de/#/home/landing |
| Services for physicians and insurance providers | 2020 | SetHealth | Medical images via iOS, android, and web app | https://set.health/ |
| Services for physicians and insurance providers | 2012 | Oskar Health | Insurance patients platform for health plans | https://www.hioscar.com/ |
| Services for physicians and insurance providers | 2016 | Bright Health | Insurance patients platform for health plans | https://brighthealthcare.com/ |
| Technology applications | 2020 | Exagol Medtech | AI for automatic venepuncture | https://excagol-medtech.com/ |
| Technology applications | 2021 | Nanostruct | Raman spectroscopy with gold particles | https://www.nanostruct.eu/en/home-english/ |
| Technology applications | 2020 | Bearcover | Robot nightworker, patients monitoring | https://www.bearcover.de/ |
| Technology applications | 2019 | Ceregate | Brain–computer interface | https://www.ceregate.com/ |

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